# MINOR IN RENEWABLE ENERGY SYSTEMS

## Rationale

The College of Engineering/ECE Department offers a minor in Renewable Energy Systems (RES). The minor aims to address the expected national, regional, and international need for personal experts in renewable energy systems for the generation, transmission, and use of energy.

## Program Objectives

The aims of the minor are to:

- Provide students with advanced expertise in renewable energy sources, systems, and policies;
- Provide students with the necessary tools to contribute to the fast-growing renewable energy systems sector;
- Prepare undergraduate students for graduate studies in renewable energy.

## Learning Outcomes

At the end of this minor, the student is expected to demonstrate the ability to:

- Use current techniques, architectures, and tools to generate, store, and distribute energy from renewable sources;
- Design systems and tools to generate, store, and distribute renewable energy.

## Program Requirements

While most relevant to electrical engineering students, this minor is offered to undergraduate RHU students in the BE or MS program, from different engineering disciplines. It may also be offered to students with a BS/BE from other universities, subject to a case-by-case evaluation of their transcripts and other specific RHU requirements.

Early in their major, interested RHU engineering students need to complete the appropriate form declaring their intent to minor in RES while completing their regular major.

To complete the RES minor, a student must

1) Declare a minor in Renewable Energy Systems by completing the Minor Declaration Form;
2) Obtain the approval of the dean of the student's college major and the dean of the College of Engineering
3) Earn a Minor Cumulative Grade Point Average of no less than 70%.
4) Complete 18 credits of coursework as specified below.

It should be noted that overlapping courses between the student's major requirements and the renewable energy systems minor requirements are counted toward fulfilling the minor requirements. The student must complete a minimum of 3 credit hours of coursework that is not counted toward the requirement for their major or any other minor.

## Career Options

The RES minor consists of six courses (18 credits), of which three are mandatory and three are electives, selected to satisfy the requirements of the proposed program objectives and learning outcomes.

## Curriculum and Program

The RES minor consists of six courses (18 credits), of which three are mandatory and three are electives, selected to satisfy the requirements of the proposed program objectives and learning outcomes.

### I. **Mandatory Courses**

The three mandatory courses (9 credits) are:

| | | |
|---|---|---|
| ELEC 320 | Electronics | 3(3,0) |
| ELEC 431 | Introduction to Renewable Energy Systems | 3(3,0) |
| ELEC 432 | Power Electronics | 3(3,0) |

### II. **Elective Courses**

Students may choose three elective courses (9 credits) from the following list.

| | | |
|---|---|---|
| ELEC 532 | Application of Power Electronics in Power Systems | 3(3,0) |
| ELEC 535 | Renewable Energy Systems for the Built Environment | 3(3,0) |
| ELEC 536 | Renewable Energy Systems | 3(3,0) |
| ELEC 542 | Advanced Power Electronics | 3(3,0) |
| ELEC 560 | Sustainable Energy | 3(3,0) |
| ELEC 561 | Wind Energy Technology | 3(3,0) |
| ELEC 562 | Solar Energy Technology | 3(3,0) |
| ELEC 563 | Smart Grid Technology | 3(3,0) |
| **Approved Electives from other engineering programs** | | |
| Civil Engineering | | |
| CIVE 506 | LEED Principles in Construction Management | 3(3,0) |
| Computer and Communication Engineering | | |
| CCEE 426 | Design of Embedded Systems | 3(3,0) |
| CCEE 564 | Machine Learning and Data Mining | 3(3,0) |
| Mechanical and Mechatronics Engineering | | |
| MECH 453 | Buildings and Energy | 3(3,0) |
| MECH 471 | Optimization in Engineering Design | 3(3,0) |

| | | |
|---|---|---|
| MECH 541 | Renewable Energy | 3(3,0) |
| MECA 581 | Systems Engineering | 3(3,0) |

## III. Courses Description

| **ELEC 320** | **Electronics** | **3(3,0)** |
|---|---|---|
| This course covers semiconductors, PN junctions, diode theory and circuits, bipolar junction transistors, transistor fundamentals, transistor biasing, amplifiers, MOSFETs, and operational amplifiers. **Prerequisite**: ELEC 210. | | |

| **ELEC 431** | **Introduction to Renewable Energy Systems** | **3(3,0)** |
|---|---|---|
| This course aims to provide students with information on renewable energy sources, including solar, wind, and wave energy, as well as biomass energy. This course will also illustrate how to link these sources to the grid and synchronize them to deliver electricity reliably and efficiently.<br>**Prerequisite**: ELE210 | | |

| **ELEC 432** | **Power Electronics** | **3(3,0)** |
|---|---|---|
| This course familiarizes the students with basic power switch technology and associated electronic circuits. In this course, power electronic circuits and switching devices are studied. Their applications in AC/DC, DC/DC, DC/AC, and AC/AC converters, as well as switching power supplies, are studied. **Prerequisite:** ELEC 320. | | |

| **ELEC 560** | **Sustainable Energy** | **3(3,0)** |
|---|---|---|
| This course is designed to provide you with the methods, tools, and perspectives to understand, critique, and ultimately influence the management of technical, economic, and policy choices regarding energy generation and use. We will focus equally on the technical, economic, political, and environmental impacts of energy. **Prerequisites**: **ELEC 431** | | |

| **ELEC 532** | **Application of Power Electronics in Power Systems** | **3(3,0)** |
|---|---|---|
| This course focuses on power electronics and its applications in power systems, covering the following topics: Flexible AC transmission systems (FACTS), conventional thyristor-controlled reactors and phase shifters, voltage source inverter (VSI)- based static condenser (STATCON), and unified power flow controller (UPFC). **Prerequisites**: ELEC 330; ELEC 432. | | |

| **ELEC 535** | **Renewable Energy Systems for the Built Environment** | **3(3,0)** |
|---|---|---|
| This course covers Energy and carbon emissions in the built environment, including data on energy consumption in Lebanon. Overview of building services – HVAC, Lighting, HWS, Lifts, and Small Power; data and benchmarks for energy consumption by end use in buildings; methods for estimating predicted energy consumption in buildings; time-based energy demand schedules; the importance of energy efficiency and the Lean-Clean-Green principle. **Prerequisite:** ELEC 210. | | |

| ELEC 536 | Renewable Energy Systems | 3(3,0) |
|---|---|---|
| This course aims to provide knowledge about energy production from renewable sources, the structure and nature of the interconnected electric power system, and the critical need for environmentally sensitive solutions. In addition, the economic and regulatory policy aspects of electricity and electricity markets are discussed. The course covers the basics of Wind energy conversion systems and microgrids with hybrid power sources. Modeling and control of renewable energy sources such as wind turbine generation, solar panels, fuel cells, and power electronics interfaces will be presented. **Prerequisite**: ELEC 431. | | |

| ELEC 542 | Advanced Power Electronics | 3(3,0) |
|---|---|---|
| This course is designed to provide students with a functional knowledge of modeling switching power converters, advanced power converter topologies, design constraints, and control methods. It also covers the operation of multi-level DC/AC inverters and matrix converters. **Prerequisite:** ELEC 432. | | |

| ELEC 561 | Wind Energy Technology | 3(3,0) |
|---|---|---|
| This course aims to provide the students with an in-depth understanding of the theoretical, technological, and economic aspects of wind energy systems. Upon successful completion of this course, students will be able to identify, assess, and select the types of wind turbines, estimate the power output of specific wind energy devices and systems, assess the structural suitability of wind towers, and evaluate the key aspects of onshore (urban) and offshore wind energy systems. **Prerequisite:** ELEC 210. | | |

| ELEC 562 | Solar Energy Technology | 3(3,0) |
|---|---|---|
| This course aims to provide students with a systematic understanding of current knowledge, problems, and insights in solar photovoltaic technologies; enable students to evaluate current research and advances in the field; and assess solar PV technologies, develop critiques, and propose solutions.<br>**Prerequisite:** ELEC 210. | | |

| ELEC 563 | Smart Grid Technology | 3(3,0) |
|---|---|---|
| This course will examine smart grid technologies and the transformational impacts of the smart grid on the industry. Students in this course will learn the fundamentals of the smart grid: its purpose and objectives, technologies, architectures, and management. Students will also learn about many of the challenges facing the smart grid as it evolves.<br>**Prerequisite:** ELEC 432 | | |